

Async & Await in JavaScript

Jab aap `await` hata dete ho, toh JavaScript server ke reply ka wait kiye bina turant agli line par chala jata hai. Woh aapko ek **Promise** return karta hai, jo basically ek "pending" ticket ya token hai ki data abhi raste mein hai.



Restaurant Analogy (Real-World Example)

Isko ek busy restaurant ki tarah samjho:

- Aapne ek burger order kiya (`fetch`).
- Cashier ne aapko ek buzzer ya token de diya (`Promise { <pending> }`).
- Agar aap `await` **nahi** karte, toh aap turant us token ko hi khane ki koshish kar rahe ho (kyunki burger toh abhi ban raha hai).
- Agar aap `await` karte ho, toh aap aaram se baithte ho, token bajne ka wait karte ho, aur fir apna asli burger (`data`) khaate ho.

1. Bina `await` Ke Kya Hota Hai?

JavaScript bahut impatient aur fast language hai. Jaise hi usko `fetch` milta hai, wo internet par request bhej deta hai. Lekin wahan se response aane mein kuch milliseconds ya seconds lagte hain. JS wait nahi karta aur turant aage badh jata hai:

```
// Request chali gayi, par data abhi aaya nahi.  
// JS ne Promise (token) pakda diya.  
const response = fetch('https://picsum.photos/v2/list');  
  
// Aap token print kar rahe ho, actual data nahi.  
console.log(response); // Output: Promise { <pending> }
```

2. `await` Ke Saath Kya Hota Hai?

`await` lagane se aap JavaScript ko strict order dete ho ki **"Jab tak task pura na ho jaye aur final data na mil jaye, is line par ruko."**

```
// JS yahan rukega jab tak server actually reply nahi karta.  
const response = await fetch('https://picsum.photos/v2/list');
```

3. Code Mein Do Baar `await` Kyun Use Hua?

Aapke code mein API se data nikalna actually ek 2-step process hota hai:

- **Step 1:** `await fetch(...)`
Yeh sirf server se connection banne aur shuruati response aane ka wait karta hai. Data abhi readable nahi hota, sirf headers aate hain.
- **Step 2:** `await response.json()`
Yeh us raw text/data ko puri tarah download karke ek proper JavaScript object (JSON format) mein convert karne ka wait karta hai, jisko aap apne code mein aasaani se use kar sako.

4. `async` Keyword Kyun Lagaya?

JavaScript ka rule hai ki aap `await` ko kisi bhi normal function ke andar aise hi use nahi kar sakte. Function ke naam ke aage `async` lagana padta hai.

`async` lagakar aap JS ko batate ho ki: *"Is function ke andar kuch aise tasks hain jo time lenge, toh jab bhi tumhe `await` dikhe, waha ruk jana."*



Bonus: Sahi Tarika (Try/Catch ke sath)

Real-world projects mein humesha errors aa sakte hain (jaise internet band hona ya API link kharab hona). Isliye humein isko `try...catch` block mein likhna chahiye, jisse agar kuch gadbad ho toh app crash na ho:

```
async function apiCalling() {
  try {
    // API ko call karo aur wait karo
    const response = await fetch('https://picsum.photos/v2/list');

    // Data ko JSON mein convert karo aur wait karo
    let data = await response.json();

    console.log(data); // Perfect data milega

  } catch (error) {
    // Agar API fail hui ya internet band hua toh yahan aayega
    console.log("Kuch gadbad ho gayi:", error);
  }
}

apiCalling();
```